Collatz Conjecture with Process Management

1. Team Members and Responsibility

As this is an individual assignment, all tasks were performed by a single member:

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2. Project Overview

This project implements the **Collatz conjecture** algorithm in C using Unix system calls. The program forks a **child process** to generate the sequence, while the **parent process** invokes `wait()` to ensure synchronized execution and proper resource cleanup.

Collatz Conjecture Logic

The Collatz conjecture concerns what happens when we take any positive integer n and apply the following algorithm:

- If n is even, n = n / 2
- If n is odd, n = 3 * n + 1

The conjecture states that all positive integers will eventually reach 1.

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3. Features

- **Process Management**: Demonstrates process creation via `fork()` and synchronization via `wait()`.
- **Isolated Execution**: Computational logic is strictly contained within the child process.
- **Robust Validation**: Ensures command-line arguments are present and are positive integers (n > 0).

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4. Compilation and Configuration Instructions

Environment

- **OS**: Developed on **macOS (Apple Silicon)**; compatible with Unix-like systems (Linux/BSD).
- **Requirements**: `gcc` or `clang` compiler.

Build the Project

Enter the project folder and use one of the following command to compile:

Method A. Using Makefile(Recommended):

```
make
```

Method B. Direclty command to complie

```
gcc collatz.c -o collatz
```

Run the Program

Run the program by passing a positive integer as a command-line argument:

```
./collatz 35
```

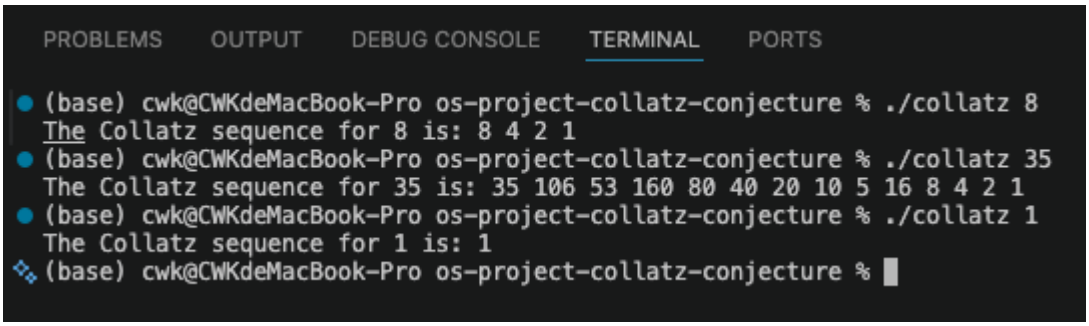
Verification

When running `./collatz 35`, the expected output:

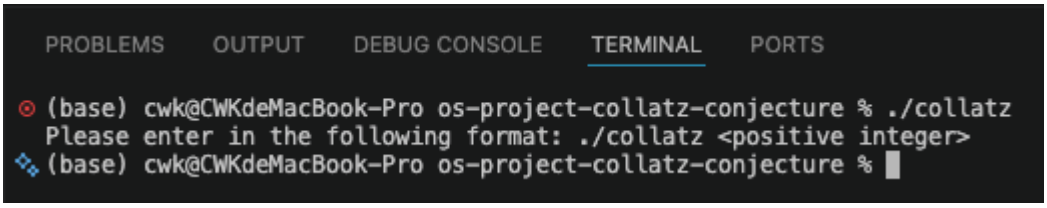
```
The Collatz sequence for 35 is: 35 106 53 160 80 40 20 10 5 16 8 4 2 1
```

5. Test results

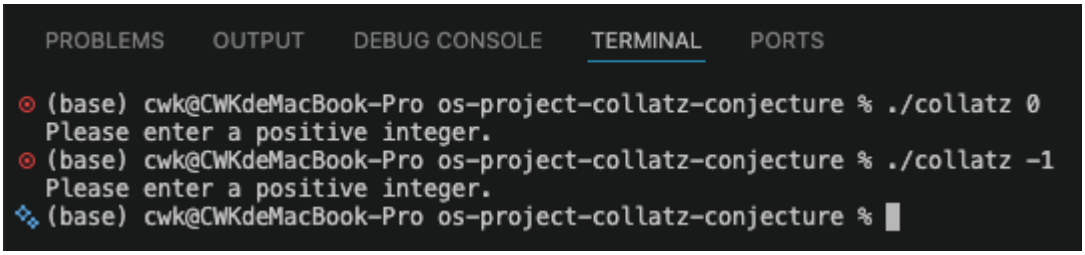
a. Basic feature(Normal Case)



b. Error handling: Missing Argument



c. Error handling: Non-postive integer



6.File Structure

- `collatz.c`: Core source code (Process logic & Collatz algorithm).
- `Makefile`: Automation script for compilation and cleanup.
- `README.md`: Project documentation.
- `snapshot/`: Directory containing execution screenshots.

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